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AIFBA EXPERIMENT : 07

```
# Install libraries
!pip install numpy pandas scikit-learn matplotlib seaborn
```

```
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (0.13.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.61.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.0)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0
```

```
# Import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, confusion_matrix, roc_auc_score
from sklearn.preprocessing import StandardScaler
from sklearn.datasets import make_classification
```

```
# Generate synthetic data
X, y = make_classification(
    n_samples=10000,
    n_features=30,
    n_informative=10,
    n_redundant=5,
    n_clusters_per_class=1,
    weights=[0.95, 0.05],
    flip_y=0,
    random_state=42
)
```

```
# Create DataFrame
data = pd.DataFrame(X, columns=[f'feature_{i}' for i in range(30)])
data['Class'] = y
# Display first rows
print(data.head())
```

```
   feature_0  feature_1  feature_2  feature_3  feature_4  feature_5  \
0  -2.101043  -2.212072  -0.332241  -7.416295   0.252392   0.253607
1   4.700580   4.709940   2.050057   5.770069   0.942296  -0.272374
2   3.445434  -6.098881   0.891359 -10.409515  -2.203294   0.808222
3   3.671999  -7.524541   1.430838  -8.827651   0.222489   0.517257
4   5.562186  -0.864840  -1.528983  -2.698827   1.223588   1.014855

   feature_6  feature_7  feature_8  feature_9  ...  feature_21  feature_22  \
0   0.644369   0.450446  -1.374330  -3.327773  ...   -0.586135    2.862179
1   0.881923   0.121683   5.728600   0.571279  ...    0.510511   -2.217313
2   0.528639   0.926021   0.760938  -4.060869  ...   -1.924878    4.042603
3   0.556610   0.132625  -1.023435  -1.237871  ...   -3.890478    1.806254
4  -0.492178   0.785472   0.029253  -0.771941  ...   -1.216911   -2.588660

   feature_23  feature_24  feature_25  feature_26  feature_27  feature_28  \
0    1.878167   -1.250302   -0.886387   -0.693075    0.293614    0.029808
1   -2.191579    2.215623   -0.796744   -0.689522    0.552602    0.249054
2    4.002704   -2.163060   -4.626833   -1.469496   -1.965914    0.117049
3    5.705049   -2.833763   -4.717457   -0.224327   -0.626313   -0.789587
```

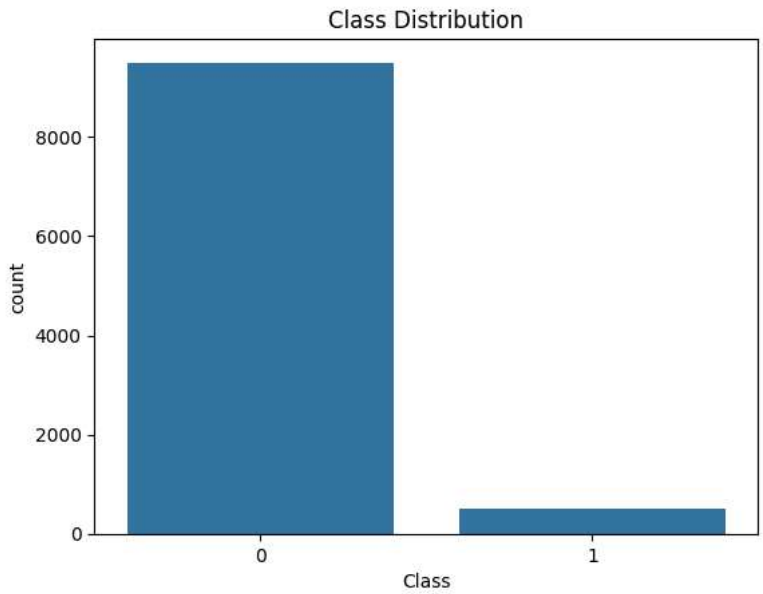
```
4    4.175046   -0.835500   -2.944726    0.848010   -0.396551    2.083104

   feature_29  Class
0    0.728677      0
1   -0.719155      0
2    0.574638      0
3    1.573872      0
4    1.057931      0

[5 rows x 31 columns]
```

```
# Check class distribution
print(data['Class'].value_counts())
# Visualize class distribution
sns.countplot(x='Class', data=data)
plt.title("Class Distribution")
plt.show()
# Scale features
scaler = StandardScaler()
features = data.drop('Class', axis=1)
scaled_features = scaler.fit_transform(features)
# Create scaled dataframe
scaled_data = pd.DataFrame(scaled_features, columns=features.columns)
scaled_data['Class'] = data['Class']
```

```
Class
0    9500
1     500
Name: count, dtype: int64
```




```
# Split dataset
X = scaled_data.drop('Class', axis=1)
y = scaled_data['Class']
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

# Train Random Forest
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)
print(model)
```

```
RandomForestClassifier(random_state=42)
```

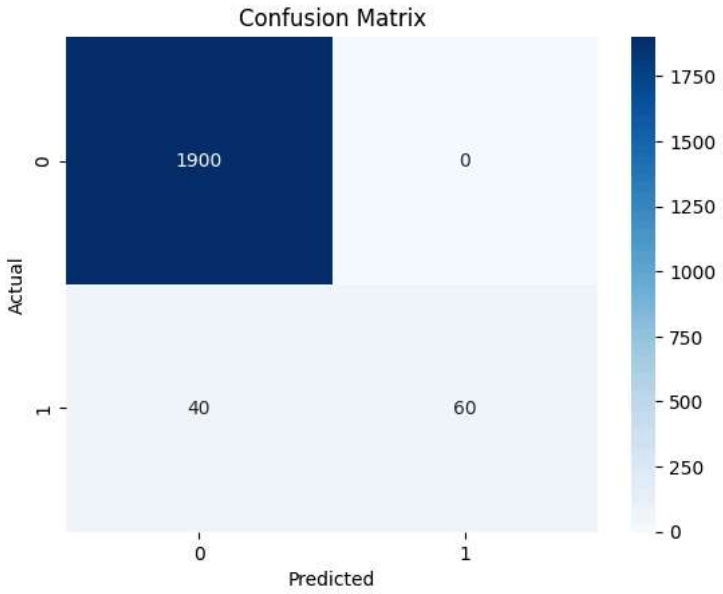
```
# Predictions
y_pred = model.predict(X_test)
# Classification report
print(classification_report(y_test, y_pred))
```

```
# Confusion matrix
conf_matrix = confusion_matrix(y_test, y_pred)
sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='Blues')
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
# ROC AUC score
roc_auc = roc_auc_score(y_test, model.predict_proba(X_test)[:,1])
print("ROC AUC Score:", roc_auc)
```

```
              precision    recall  f1-score   support

           0       0.98      1.00      0.99      1900
           1       1.00      0.60      0.75       100

    accuracy                           0.98      2000
   macro avg       0.99      0.80      0.87      2000
weighted avg       0.98      0.98      0.98      2000
```




```
ROC AUC Score: 0.9916184210526315
```